



Medical Microbiology

Lecture Nine

Streptococcus mutans

Third stage- College of Dentistry

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Introduction

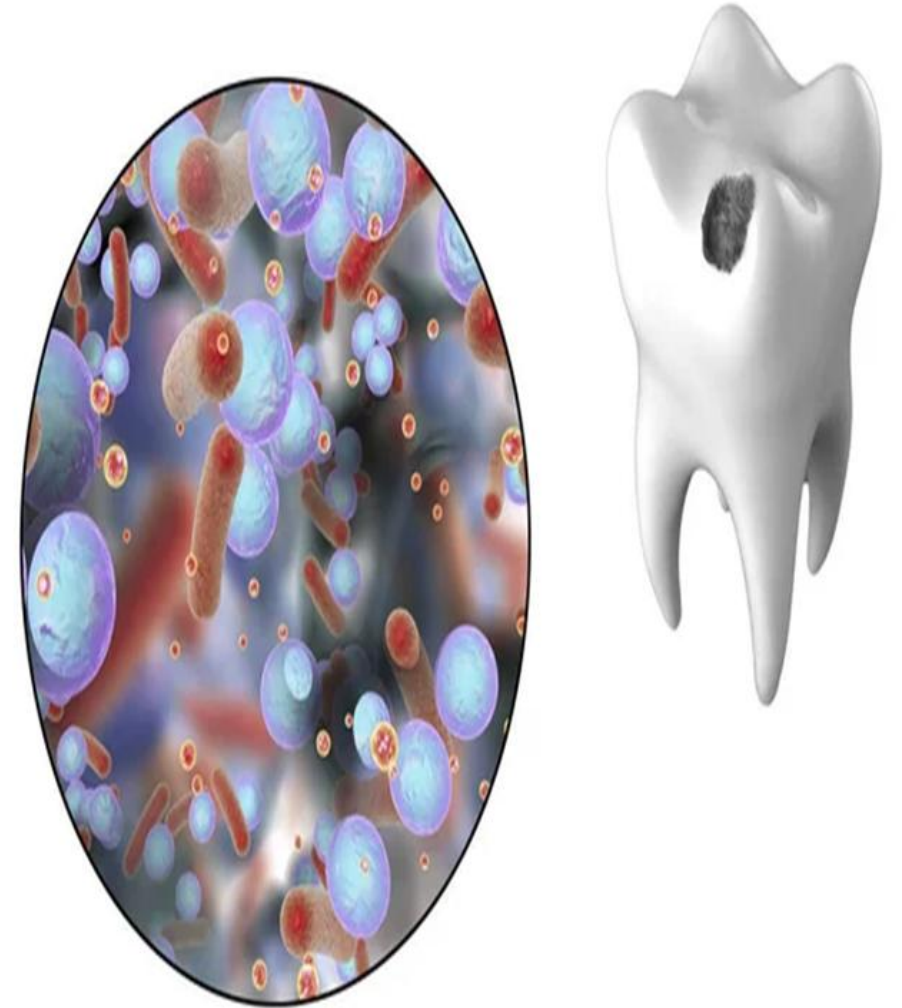
Streptococcus: is a genus of spherical, gram-positive bacteria that are commonly found in various environments, including the human body.

- **Natural Habitat**

- Part of the **oral normal flora**, especially hard surfaces of the teeth.
- Colonizes pits, fissures, and gingival margins.
- Appears early after tooth eruption.

Streptococcus mutans:

- It is a Gram-positive, facultative anaerobic coccus and one of the primary bacteria responsible for **dental caries**.
- It belongs to the **viridans group streptococci** and is a natural part of the oral microbiome.



Why it causes dental caries?

Because the following:

1-Adhesion:

Produces extracellular polysaccharides that allow strong attachment to tooth enamel and formation of **dental biofilm (plaque)**.

2.Colonization:

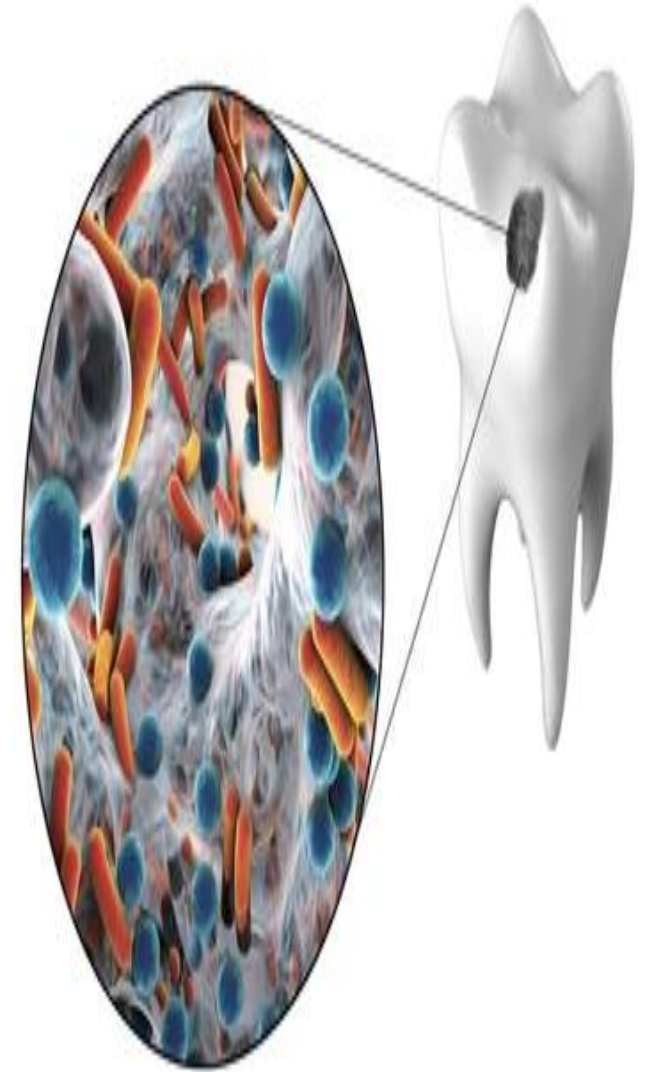
Binds to salivary pellicle using adhesions and glucans. Often acquired in early childhood, typically from caregivers.

• Risk Factors:

High sugar diet.

Poor oral hygiene.

Reduced salivary flow or xerostomia.

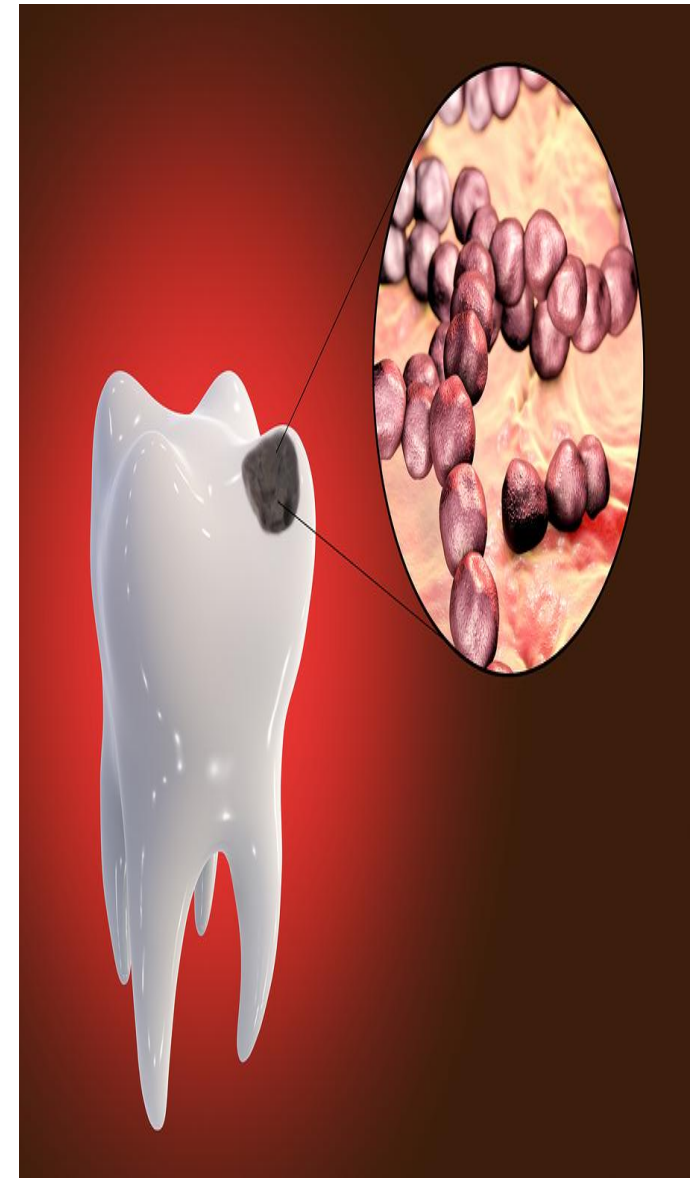


3-Fermentation of sugars:

Ferments carbohydrates especially sucrose and produces **lactic acid**, which lowers pH below the critical level (5.5), leading to **demineralization**, white-spot lesions, and eventually **cavitation**.

4-Acid tolerance:

Can survive and remain active in acidic environments, giving it an advantage over other bacteria.



Morphology :

- **Appears** as spherical or oval-shaped G+ve bacteria under the microscope.
- **Non-motile:** Does not have flagella for movement.
- **Facultative anaerobe:** Can survive in both aerobic and anaerobic conditions.



Classification: Streptococci are classified into groups based on their **hemolytic** properties on blood agar:

1-Alpha-hemolytic: Produces partial hemolysis on blood agar, resulting in a greenish discoloration.

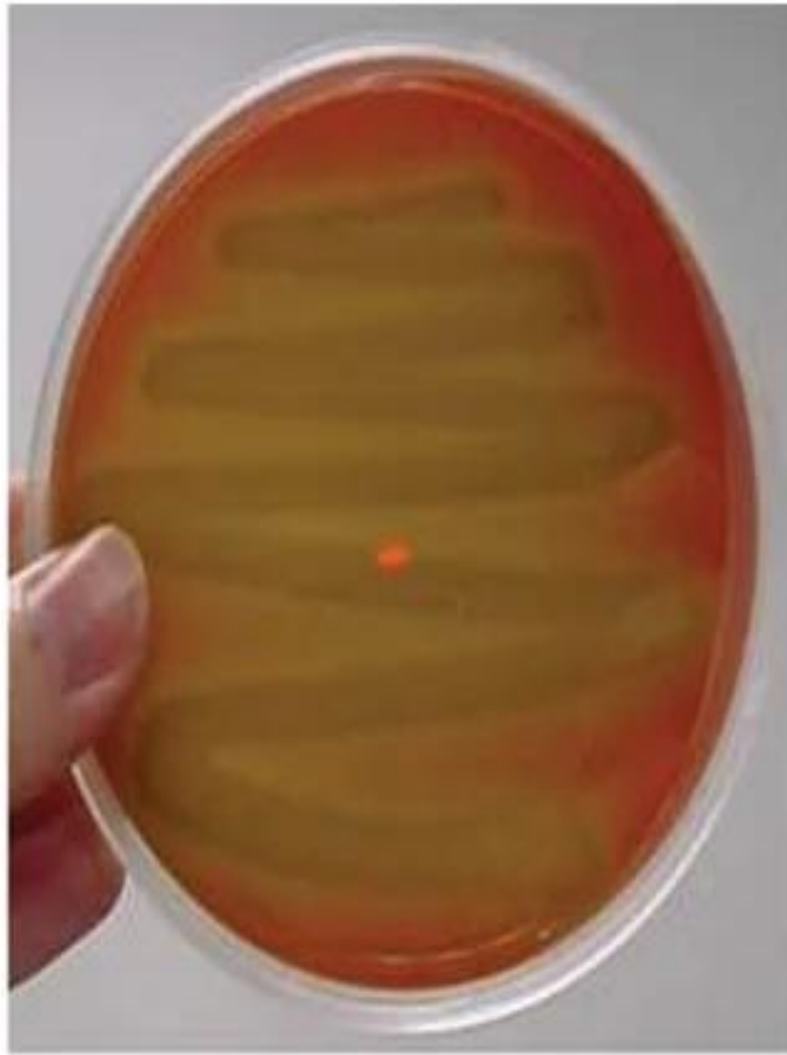
(e.g., *Streptococcus mutans* and *St. pneumonia*).

2-Beta-hemolytic: Completely lyse red blood cells on blood agar (e.g., *Streptococcus pyogenes*).

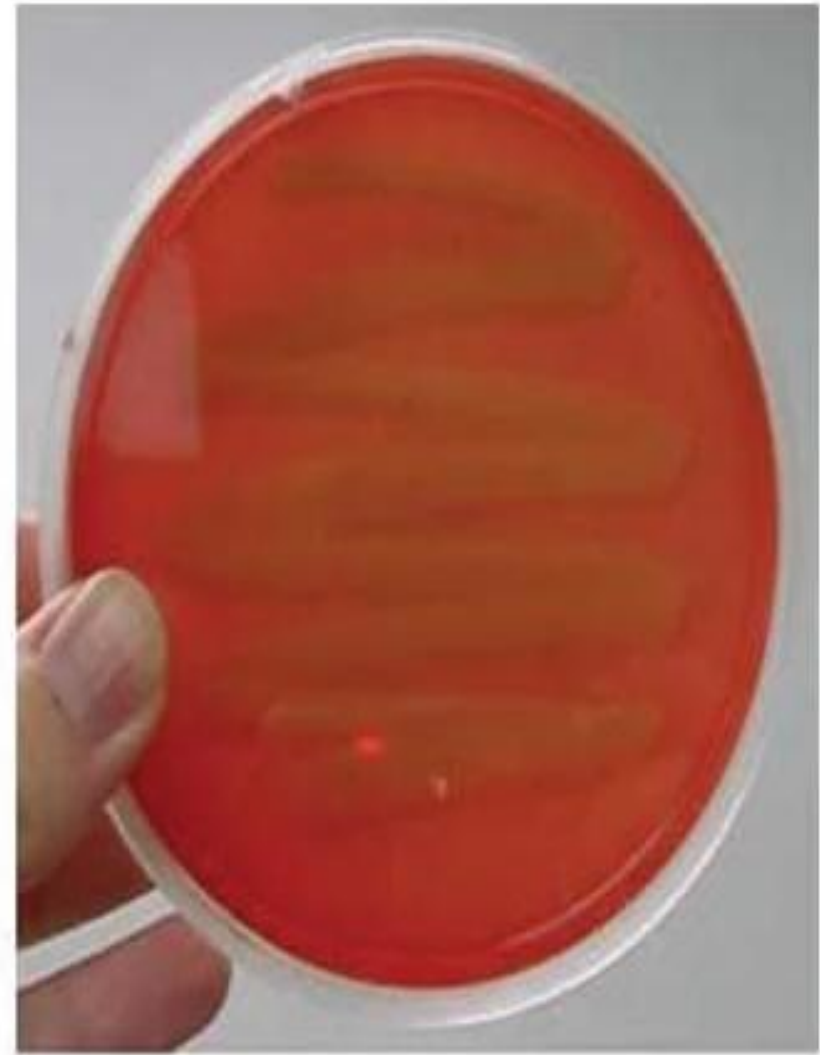
3-Gamma-hemolytic: No hemolysis
(e.g., *Enterococcus species*).



Beta Hemolysis



Alpha Hemolysis



Gamma Hemolysis

Virulence Factors:

1-Adhesion:

Produces surface proteins that facilitate attachment to tooth surfaces. Key adhesions include **Surface Protein Antigen P (SpaP)** or **(Antigen I/II)** (It is a major surface protein that plays a crucial role in the adhesion of *S. mutans* to the tooth surface and is considered one of its most important virulence factors) and **glucan-binding proteins (GBPs)** (These are proteins that bind to glucans (sticky polysaccharides) produced by the bacteria from sucrose).

2-Acid Production:

Ferments dietary carbohydrates to produce lactic acid, lowering the local pH and Contributes to enamel demineralization and cavity formation.

3-Production of Bacteriocins:

Releases antimicrobial peptides to inhibit competing oral microbiota, enhancing its dominance in the biofilm.



4-Biofilm Formation:

Synthesizes extracellular polysaccharides (EPS) using glucosyl-transferases (GTFs).

Forms a sticky biofilm (dental plaque) that adheres to teeth and protects the bacteria from external challenges.



Laboratory Diagnosis:

1. Sample Collection

- Dental plaque scrapings
- Swabs from tooth surfaces

2. Microscopy:

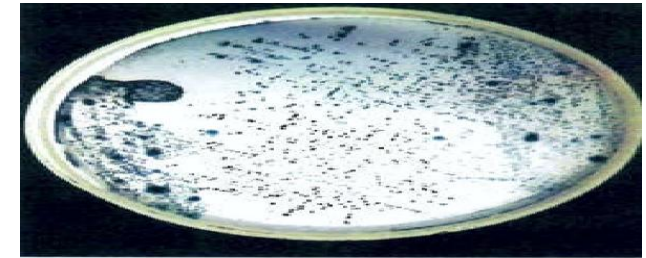
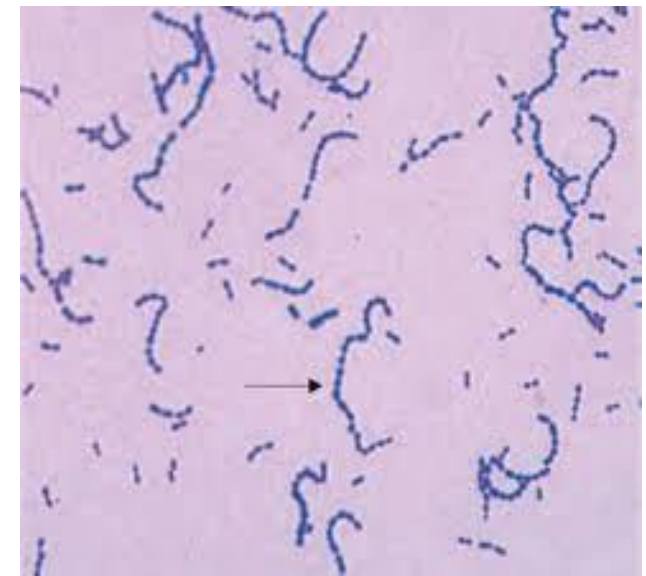
Identification of Gram positive cocci in chains.

3. Culture Media:

Grows on **Mitis-Salivarius** agar supplemented with sucrose, Produces raised, frosted-glass colonies.

4. Molecular Methods:

PCR-based detection of *S. mutans* specific genes, such as GTFB or spaP.



5. Biochemical Tests:

- **Sugar fermentation:** ferments glucose, sucrose, mannitol.
- **Hydrolyzes sucrose** to produce extracellular glucans (key virulence).
- **Produces** lactic acid.
- **Catalase test:** negative
- **Bile esculin test:** negative
- **Grows on MSB agar (Mitis-Salivarius Bacitracin)** selective medium for *S. mutans*.

Prevention and Control:

1. Oral hygiene measures

- Tooth brushing with fluoride toothpaste.
- Flossing to disrupt biofilm.

2. Fluoride therapy

- Enhances enamel resistance.
- Inhibits bacterial metabolism.
- Promotes remineralization.

3. Dietary control

- Reduce frequency of sucrose intake.
- Avoid sugary snacks between meals.

4. Antimicrobial agents

5. Probiotic therapy

- Certain *Lactobacillus* strains reduce *S. mutans* colonization.

Treatment:

1. Caries management

- Removal of decayed tissue
- Restoration with composite/amalgam
- Enamel remineralization therapy (fluoride)

2. Dental Interventions:

Removal of carious lesions and restoration with dental fillings.

3. Antimicrobial control

- Chlorhexidine 0.12%
- Fluoride varnish
- Xylitol (chewing gum)





Any Question?



Thank you